Homework 1

Graded

Student

Colin Skinner

Total Points

100 / 100 pts

Question 1

Problem 1

15 / 15 pts

✔ **– 0 pts** Correct

– 5 pts Projection Error

– 1 pt Incorrect / Incomplete Final Answer

Question 2

Problem 2

10 / 10 pts

✔ **– 0 pts** Correct

– 2 pts Incomplete / Incorrect Final Answer

– 1 pt Minor Calculation Error

Question 3

Problem 3

25 / 25 pts

3.1 **a**

12 / 12 pts

✔ **– 0 pts** Correct

– 1 pt Sign / Minor Math Error

3.2 **b**

8 / 8 pts

✔ **– 0 pts** Correct

– 0 pts Error based on part a

3.3 **c**

5 / 5 pts

✔ **– 0 pts** Correct

– 1 pt failed to include momentum change as a result of velocity change $m\dot{v}$

– 2 pts included mass as a function of time but failed to include momentum change as a result of the mass change $\dot{m}v$

– 4 pts Incorrect formulation of the equation of motion

Question 4

Problem 4 — **15** / 15 pts

4.1 **a** — **5** / 5 pts

✔ **– 0 pts** Correct

– 1 pt Small error in initial vector values

4.2 **b** — **5** / 5 pts

✔ **– 0 pts** Correct

– 2.5 pts Incorrect formula for angle between 2 vectors

– 1 pt Error in intial vector values

4.3 **c** — **5** / 5 pts

✔ **– 0 pts** Correct

– 0.5 pts Poor plot resolution/insufficient contour lines

– 2.5 pts Incorrect equation for Z resulting in incorrect plot

– 1.5 pts Incorrect gradient vectors

– 1 pt Incorrect bounds

Question 5

Problem 5 — **20** / 20 pts

5.1 **a** — **9** / 9 pts

✔ **– 0 pts** Correct

– 1.5 pts Missing plot of x1+x2

– 1.5 pts slightly incorrect plots

– 3 pts Incorrect plot

– 5 pts Missing Plot

5.2 **b** — **11** / 11 pts

✔ **– 0 pts** Correct

– 1 pt Missing some explanation key points or slightly incorrect: SHM, unchanging shape with different IC's, etc

– 1.5 pts Missing x1+x2

– 2 pts Insufficient/incorrect analysis of behavior

– 2 pts Some errors with plot

– 3 pts Incomplete/Incorrect Plot

Question 6

Problem 6

15 / 15 pts

- ✔ **+ 8 pts** Description of System
- ✔ **+ 7 pts** Draws a picture

+ 6 pts Insufficient description of a specific problem

Question assigned to the following page: 1

following unit vector triad:

$e_1 = i; \quad e_2 = (i + j)/\sqrt{2}; \quad e_3 = (i + k)/\sqrt{2}.$

$e_1 = i \quad e_2 = \frac{\hat{i}}{\sqrt{2}} + \frac{\hat{j}}{\sqrt{2}} \quad e_3 = \frac{\hat{i}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}}$

$A = A_1 e_1 + A_2 e_2 + A_3 e_3 = A_1 i + \frac{A_2 \hat{i}}{\sqrt{2}} + \frac{A_2 \hat{j}}{\sqrt{2}} + \frac{A_3 \hat{i}}{\sqrt{2}} + \frac{A_3 \hat{k}}{\sqrt{2}}$

$\frac{A_3 \hat{k}}{\sqrt{2}} = A_z \hat{k} \qquad A_3 = \sqrt{2} A_z$

$\frac{A_2 \hat{j}}{\sqrt{2}} = A_y \hat{j} \qquad A_2 = \sqrt{2} A_y$

$A_1 \hat{i} + \frac{A_2}{\sqrt{2}} \hat{i} + \frac{A_3}{\sqrt{2}} \hat{i} = A_x \hat{i}$

$A_1 + \frac{1}{\sqrt{2}}(\sqrt{2} A_y) + \frac{1}{\sqrt{2}}(\sqrt{2} A_z) = A_x \qquad A_1 = A_x - A_y - A_z$

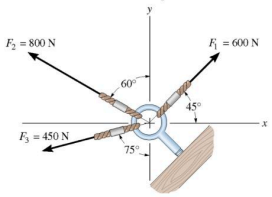
$A_1 = A_x - A_y - A_z$

$A_2 = \sqrt{2} A_y$

$A_3 = \sqrt{2} A_z$

Question assigned to the following page: 2

counterclockwise from the positive x axis.



X-dir

$F_x = F_1 \cos 45° - F_2 \sin 60° - F_3 \sin 75°$

$= (600N)\cos 45° - (800N)\sin 60° - (450N)\sin 75°$ $\quad F_x = -703.22\,N$

y-dir

$F_y = F_1 \sin 45° + F_2 \cos 60° - F_3 \cos 75°$

$= (600N)\sin 45° + (800N)\cos 60° - (450N)\cos 75°$ $\quad F_y = 707.80\,N$

$F_R = \sqrt{F_x^2 + F_y^2} = \sqrt{(-703.22N)^2 + (707.8N)^2}$

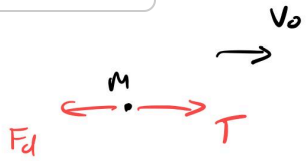
$F_R = 997.75$

$\theta = \tan^{-1}\left(\frac{F_y}{F_x}\right) = \tan^{-1}\left(\frac{707.8N}{-703.22N}\right) = -45.1° + 180°$ Quadrant 3

$\theta = 135°$

Questions assigned to the following page: 3.1 and 3.2

a. Find the speed of the airplane as a function of time.
b. Find the eventual steady-state speed of the airplane.
c. If this were a rocket plane (like the Bell X-1) and you had to consider the mass loss of fuel, how would you change your formulation of the problem? (You don't have to actually solve this problem unless you really want to. Just describe your approach.)



a) $\Sigma F = ma \rightarrow T - F_d = T - bv^2 = ma = m\frac{dv}{dt}$

$$\frac{dv}{dt} = \frac{1}{m}(T - bv^2) \Rightarrow \int_{v_0}^{v} \frac{1}{T - bv^2}\,dv = \int_0^t \frac{1}{m}\,dt$$

$$\int_{v_0}^{v} \frac{1}{T - bv^2}\,dv = \frac{1}{T}\int_{v_0}^{v} \frac{1}{1 - \frac{b}{T}v^2}\,dv = \frac{1}{\sqrt{bT}}\int_{\sqrt{\frac{b}{T}}v_0}^{\sqrt{\frac{b}{T}}v} \frac{1}{1-u^2}\,du = \frac{1}{\sqrt{bT}}\tanh^{-1}(u)\Big|_{\sqrt{\frac{b}{T}}v_0}^{\sqrt{\frac{b}{T}}v}$$

$u = \sqrt{\frac{b}{T}}v$

$du = \sqrt{\frac{b}{T}}\,dv$

$$= \frac{1}{\sqrt{bT}}\left[\tanh^{-1}\left(\sqrt{\tfrac{b}{T}}v\right) - \tanh^{-1}\left(\sqrt{\tfrac{b}{T}}v_0\right)\right]$$

$$\frac{1}{\sqrt{bT}}\left[\tanh^{-1}\left(\sqrt{\tfrac{b}{T}}v\right) - \tanh^{-1}\left(\sqrt{\tfrac{b}{T}}v_0\right)\right] = \frac{t}{m}$$

$$\tanh^{-1}\left(\sqrt{\tfrac{b}{T}}v\right) - \tanh^{-1}\left(\sqrt{\tfrac{b}{T}}v_0\right) = \frac{\sqrt{bT}}{m}t$$

$$\tanh^{-1}\left(\sqrt{\tfrac{b}{T}}v\right) = \frac{\sqrt{bT}}{m}t + \tanh^{-1}\left(\sqrt{\tfrac{b}{T}}v_0\right)$$

$$\sqrt{\tfrac{b}{T}}v = \tanh\left(\frac{\sqrt{bT}}{m}t + \tanh^{-1}\left(\sqrt{\tfrac{b}{T}}v_0\right)\right)$$

$$\boxed{v = \sqrt{\frac{T}{b}}\tanh\left(\frac{\sqrt{bT}}{m}t + \tanh^{-1}\left(\sqrt{\tfrac{b}{T}}v_0\right)\right)}$$

b) steady state when $a = 0$ so $T - bv^2 = 0$

$$T = bv^2 \qquad v^2 = \frac{T}{b} \rightarrow \boxed{v = \sqrt{\frac{T}{b}}}$$

Question assigned to the following page: 3.3

terms of $F = \frac{dp}{dt}$ since mass is no longer constant.

So $T - bv^2 = \frac{d}{dt}(mv) = m\frac{dv}{dt} + v\frac{dm}{dt}$

$$T - bv^2 = m\frac{dv}{dt} + v\dot{m}$$

Question assigned to the following page: 4.3

problems in this class. Use MATLAB's *publish* feature on newer versions to export to PDF. Answers should be clearly indicated; this problem should be solved in MATLAB (even if it is easy for you to do by hand).

a. Find the unit vector in the direction of (1, -2, 4).

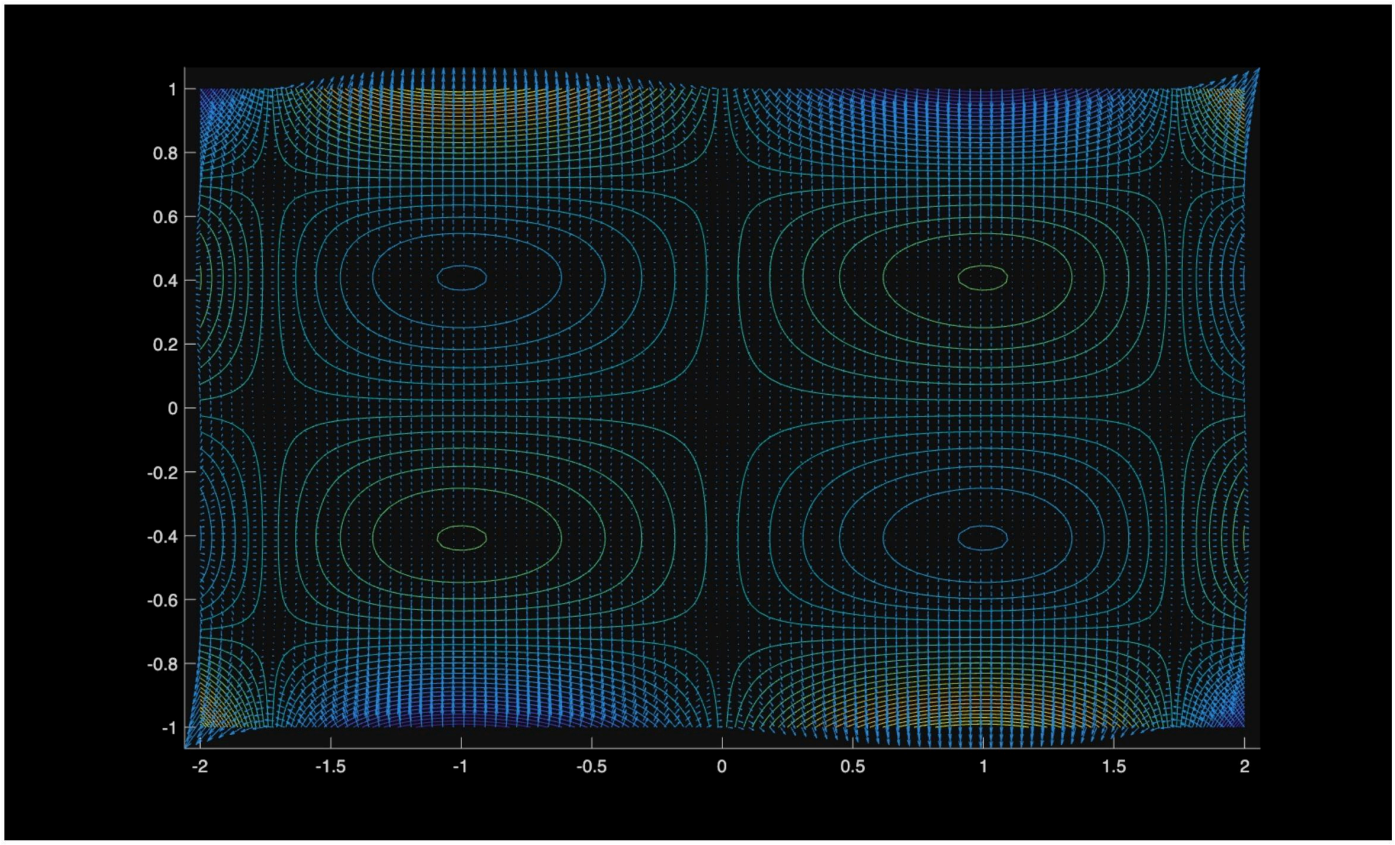
b. Find the angle (in radians) between the vectors (1, -2, 3) and (3, 1, 4).

c. Generate a plot of the level curves for the function

$$f(x,y) = (x^3 - 3x)(2y^3 - y)$$

for $-2 \le x \le 2$ and $-1 \le y \le 1$. On your plot of the level curves, draw vectors in the direction of the gradient of f.

Hint: look into MATLAB's *contourf* and *quiver* functions for plotting.



Questions assigned to the following page: 4.1, 4.2, and 4.3

```
clc
clear all
close all

unit_vector = sprintf('%.4f ', [1 -2 4] / norm([1 -2 4]));
fprintf("4.1: Unit vector = [ %s]\n\n", unit_vector)

v1 = [1 -2 3];
v2 = [3 1 4];

angle_between = acos( dot(v1, v2) / norm(v1) / norm(v2));
fprintf("4.2: Angle between = %.2f rad\n\n", angle_between);

x = linspace(-2, 2, 100);
y = linspace(-1, 1, 100);
[X,Y] = meshgrid(x,y);

Z = (X.^3 - 3*X) .* (2*Y.^3 - Y);

U = (3*X.^2 - 3) .* (2*Y.^3 - Y);
V = (X.^3 - 3*X) .* (6*Y.^2 - 1);
figure(1)
hold on
contour(X, Y, Z, 40)
quiver(X, Y, U, V, 2)
hold off
saveas(gcf, "contours.png")
```

```
4.1: Unit vector = [ 0.2182 -0.4364 0.8729 ]

4.2: Angle between = 0.82 rad
```

Questions assigned to the following page: 5.1 and 5.2

Using ODE45 or ODE113, numerically integrate the following set of equations:

$$\ddot{x}_1 = -ax_1 + b(x_2 - x_1)$$
$$\ddot{x}_2 = -c(x_2 - x_1)$$

where the dots indicate time derivatives (this is typically the set of equations one arrives at for coupled oscillating systems, like a set of masses and springs). Solve these equations from t=0 to t=100 for the following initial conditions:

$$x_1(0) = 1, \dot{x}_1(0) = 0$$
$$x_2(0) = 1, \dot{x}_2(0) = 0$$

Solve the equations for 2 cases and plot x_1, x_2 and x_1+x_2 vs. time:

a. a=4, b=1, c=7.5

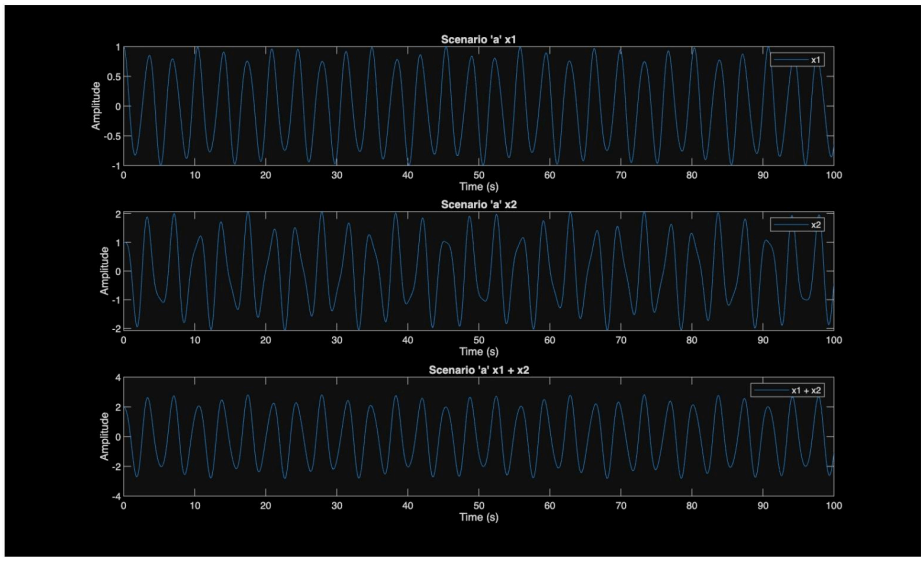
b. a=1, b=1, c=1.5

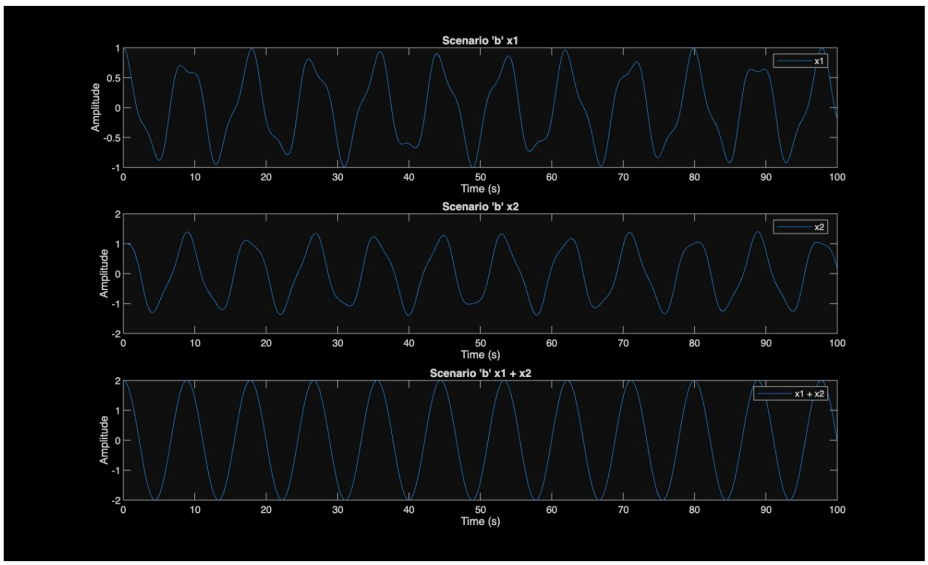
Notice anything interesting in the second case? Try changing initial conditions for this case. What happens?

$$y_1 = \dot{x}_1 \qquad \dot{y}_1 = -ax_1 + b(x_2 - x_1)$$

$$y_2 = \dot{x}_2 \qquad \dot{y}_2 = -c(x_2 - x_1)$$

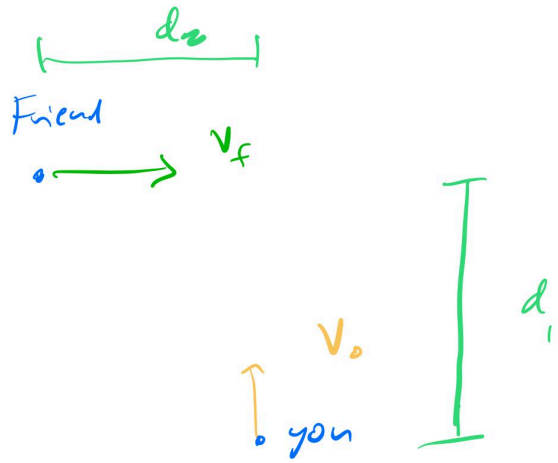
$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix} \qquad \dot{\vec{x}} = \begin{bmatrix} y_1 \\ y_2 \\ -ax_1 + b(x_2 - x_1) \\ -c(x_2 - x_1) \end{bmatrix}$$





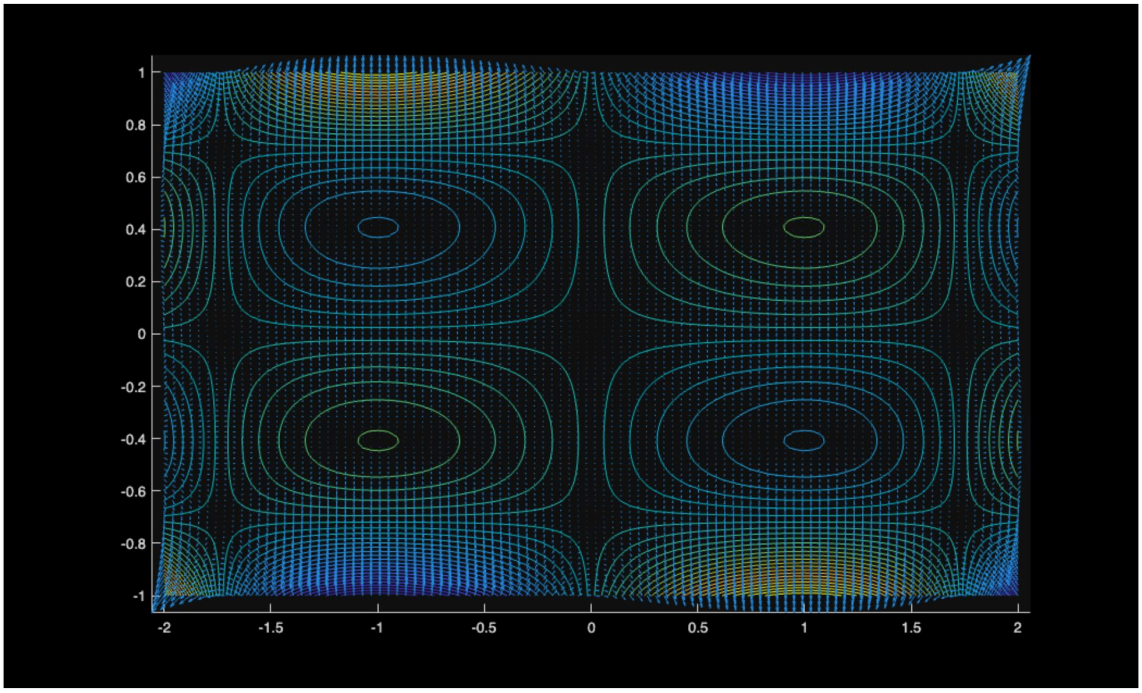
Question assigned to the following page: 6

ight experiment" where you will identify and solve a "Classical Dynamics" problem of your choosing. For this week, describe in words and draw a picture of a dynamics problem that you recognize right now in your immediate vicinity (e.g. your desk, the book on your desk, a scene outside the nearest window). Be creative!



A friend and I are biking past each other on campus. He is traveling at the speed v_f = 4 m/s and I am traveling at the speed 7 m/s. I am at a perpendicular distance of 14 m from his path of travel, and he is 18 meters from mine. At what point are we the closest, and at what angle should my friend wave to me clockwise from his direction of travel?

Questions assigned to the following page: 4.3 and 5.1



ODE

```
function [x_dot] = odefun(t, x, a, b, c)
    x1 = x(1);
    x2 = x(2);
    y1 = x(3);
    y2 = x(4);

    dx1 = y1;
    dx2 = y2;
    dy1 = -a * x1 + b * (x2 - x1);
    dy2 = -c * (x2 - x1);

    x_dot = [dx1; dx2; dy1; dy2];
end

initial_conditions = [1; 1; 0; 0];

tspan = [0, 100];

[t1,y1] = ode45(@(t,y) odefun(t, y, 4, 1, 7.5), tspan, initial_conditions);
[t2,y2] = ode45(@(t,y) odefun(t, y, 1, 1, 1.5), tspan, initial_conditions);

figure(2)

subplot(3,3,1:3)
plot(t1, y1(:,1))
title("Scenario 'a' x1")
```

Question assigned to the following page: 5.1

```
legend('x1')
xlabel("Time (s)")
ylabel("Amplitude")

subplot(3,3,4:6)
plot(t1, y1(:,2))
title("Scenario 'a' x2")
legend('x2')
xlabel("Time (s)")
ylabel("Amplitude")

subplot(3,3,7:9)
plot(t1, y1(:,1) + y1(:,2))
title("Scenario 'a' x1 + x2")
legend('x1 + x2')
xlabel("Time (s)")
ylabel("Amplitude")

saveas(gcf, "scenario_a.png")

figure(3)

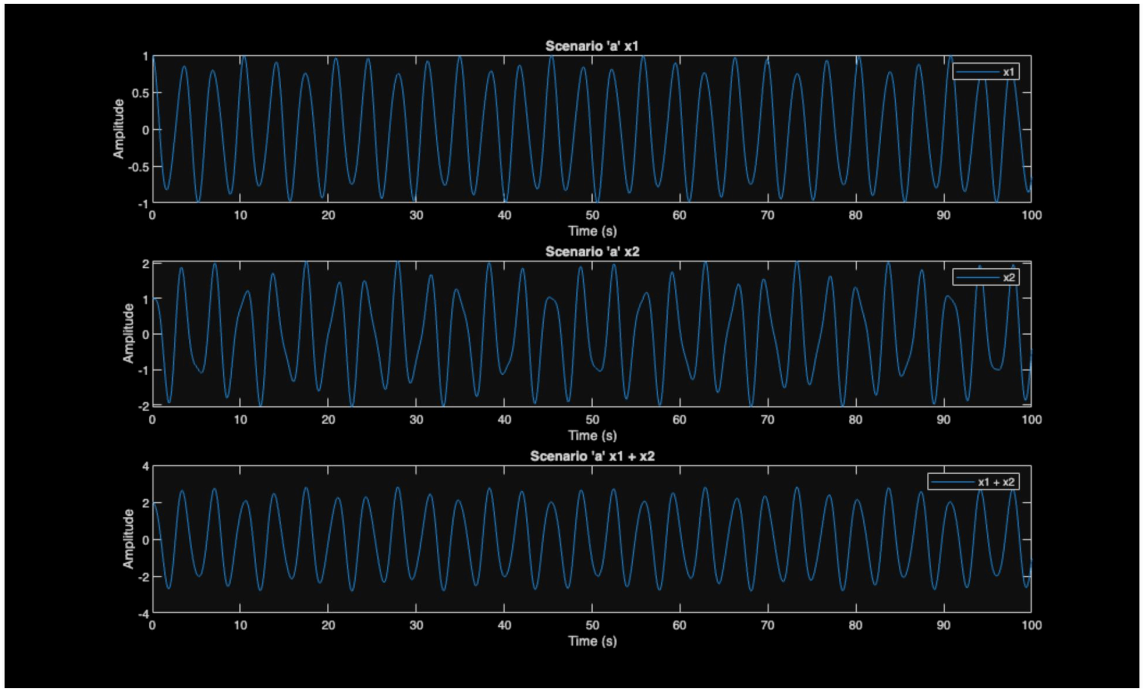
subplot(3,3,1:3)
plot(t2, y2(:,1))
title("Scenario 'b' x1")
legend('x1')
xlabel("Time (s)")
ylabel("Amplitude")

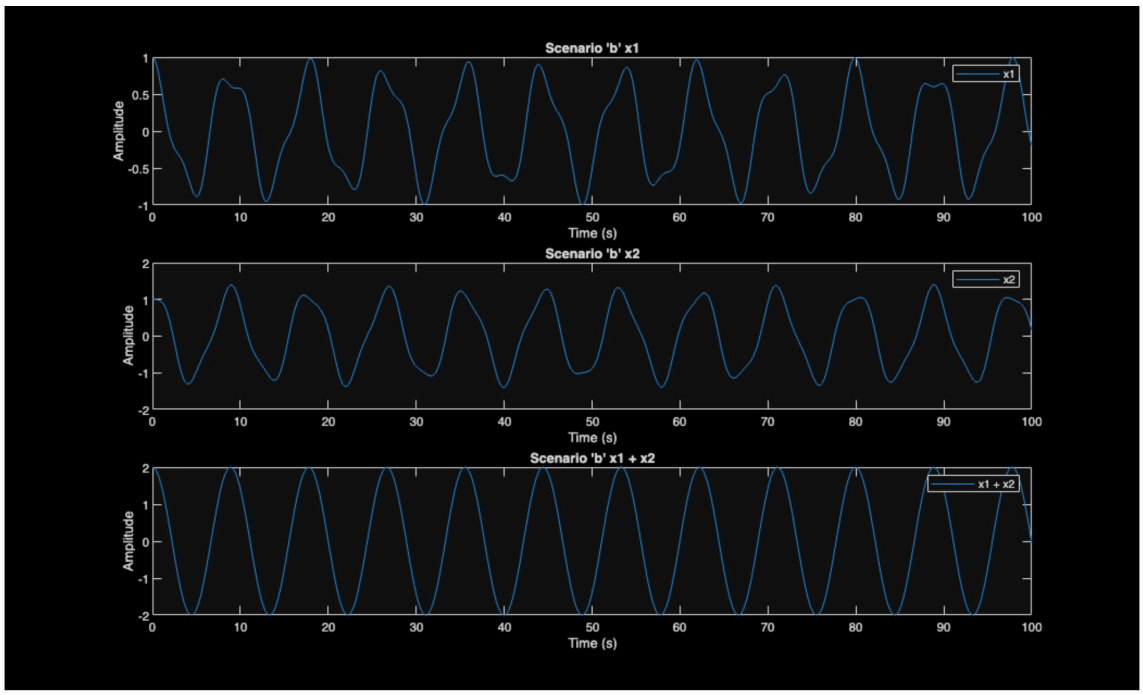
subplot(3,3,4:6)
plot(t2, y2(:,2))
title("Scenario 'b' x2")
legend('x2')
xlabel("Time (s)")
ylabel("Amplitude")

subplot(3,3,7:9)
plot(t2, y2(:,1) + y2(:,2))
title("Scenario 'b' x1 + x2")
legend('x1 + x2')
xlabel("Time (s)")
ylabel("Amplitude")

saveas(gcf, "scenario_b.png")
```

Question assigned to the following page: 5.1





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